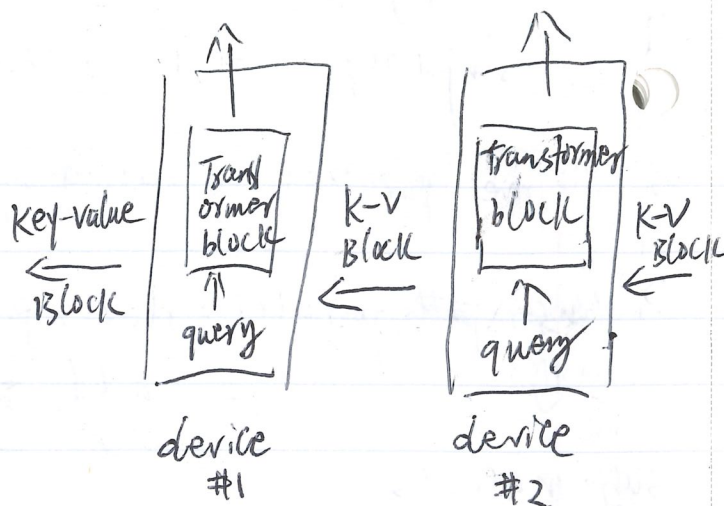


Other parallel strategies:

① Context parallel / Ring attention
(split activations across GPUs in a long sequence)



② $\rightarrow$ MLPs
experts parallel:

$\Rightarrow$ split experts across GPUs
(large MLPs to small MLPs)

$\Leftrightarrow$ computation/communication overlap

$\Rightarrow$ sparsity: only a subset of experts is activated per input token.

Recap: LLM parallelism table

	sync overhead	memory	Bandwidth	Batch size	ease of use
DDP/ZeRO 1	per-batch.	No scaling	$2 \times \# \text{param}$	linear	very
FSDP/ZeRO 3	$3 \times$ per-FSDP block	linear.	$3 \times \# \text{param}$	linear	very
Pipeline	per-pipeline	Linear.	Activations	linear	very No
Tensor + seq	$2 \times$ transformer block	Linear	$8 \times$ activations per-layer all-reduce	linear	No